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1 # nombre: Merin Zepeda Esteban Gabriel
2 # grupo: 951 fecha: 1 de mayo del 2026
3
4 import mysql.connector
5
6
7 class conexion:
8     def __init__(self, host, user, password, database
9     ):
10         self.host = host
11         self.user = user
12         self.password = password
13         self.database = database
14         self.conexion = None
15
16     def conectar(self):
17         self.conexion = mysql.connector.connect(
18             host=self.host,
19             user=self.user,
20             password=self.password,
21             database=self.database
22         )
23         return self.conexion
24
25     def desconectar(self):
26         if self.conexion:
27             self.conexion.close()
28
29 class sqlPais(conexion):
30     def __init__(self, host, user, password, database
31     ):
32         super().__init__(host, user, password,
33         database)
34
35     def insertar(self, id, nombre):
36         try:
37             con = self.conectar()
38             cursor = con.cursor()
39             cursor.execute("insert into pais (id,
40             nombre) values (%s, %s)", (id, nombre))
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38         con.commit()
39         return True
40     except:
41         return False
42     finally:
43         self.desconectar()
44
45     def editar(self, id, nombre_nuevo):
46         try:
47             con = self.conectar()
48             cursor = con.cursor()
49             cursor.execute("select id from pais where
nombre = %s", (nombre_nuevo,))
50             if cursor.fetchone():
51                 return False
52             cursor.execute("update pais set nombre
= %s where id = %s", (nombre_nuevo, id))
53             con.commit()
54             return True
55         except:
56             return False
57         finally:
58             self.desconectar()
59
60     def eliminar(self, id):
61         try:
62             con = self.conectar()
63             cursor = con.cursor()
64             cursor.execute("delete from pais where id
= %s", (id,))
65             con.commit()
66             return cursor.rowcount > 0
67         except:
68             return False
69         finally:
70             self.desconectar()
71
72     def consultar(self, filter):
73         try:
74             con = self.conectar()
75             cursor = con.cursor()
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76         cursor.execute(f"select * from pais
    where {filter}")
77         return cursor.fetchall()
78     except:
79         return []
80     finally:
81         self.desconectar()
82
83
84 class sqlOlimpiada(conexion):
85     def __init__(self, host, user, password,
    database):
86         super().__init__(host, user, password,
    database)
87
88     def insertar(self, id, year):
89         try:
90             con = self.conectar()
91             cursor = con.cursor()
92             cursor.execute("insert into olimpiada (
    id, year_olimpiada) values (%s, %s)", (id, year))
93             con.commit()
94             return True
95         except:
96             return False
97         finally:
98             self.desconectar()
99
100    def editar(self, id, year_nuevo):
101        try:
102            con = self.conectar()
103            cursor = con.cursor()
104            cursor.execute("select id from olimpiada
    where year_olimpiada = %s", (year_nuevo,))
105            if cursor.fetchone():
106                return False
107            cursor.execute("update olimpiada set
    year_olimpiada = %s where id = %s", (year_nuevo, id
    ))
108            con.commit()
109            return True

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110         except:
111             return False
112         finally:
113             self.desconectar()
114
115     def eliminar(self, id):
116         try:
117             con = self.conectar()
118             cursor = con.cursor()
119             cursor.execute("delete from olimpiada
120 where id = %s", (id,))
121             con.commit()
122             return cursor.rowcount > 0
123         except:
124             return False
125         finally:
126             self.desconectar()
127
128     def consultar(self, filter):
129         try:
130             con = self.conectar()
131             cursor = con.cursor()
132             cursor.execute(f"select * from olimpiada
133 where {filter}")
134             return cursor.fetchall()
135         except:
136             return []
137         finally:
138             self.desconectar()
139
140 class ResultadosSQL(conexion):
141     def __init__(self, host, user, password,
142 database):
143         super().__init__(host, user, password,
144 database)
145
146     def insertar(self, idOlimpiada, idPais, idGenero
147 , oro, plata, bronce):
148         try:
149             con = self.conectar()
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146         cursor = con.cursor()
147         cursor.execute(
148             "insert into resultados (idOlimpiada
149             , idPais, idGenero, oro, plata, bronce) values (%s
150             , %s, %s, %s, %s, %s)",
151             (idOlimpiada, idPais, idGenero, oro
152             , plata, bronce)
153         )
154         con.commit()
155         return True
156     except:
157         return False
158     finally:
159         self.desconectar()
160
161     def editar(self, idOlimpiada, idPais, idGenero,
162     oro, plata, bronce):
163         try:
164             if not all(isinstance(v, int) and v > 0
165             for v in [oro, plata, bronce]):
166                 return False
167             con = self.conectar()
168             cursor = con.cursor()
169             cursor.execute(
170                 "update resultados set oro = %s,
171                 plata = %s, bronce = %s where idOlimpiada = %s and
172                 idPais = %s and idGenero = %s",
173                 (oro, plata, bronce, idOlimpiada,
174                 idPais, idGenero)
175             )
176             con.commit()
177             return cursor.rowcount > 0
178         except:
179             return False
180         finally:
181             self.desconectar()
182
183     def eliminar(self, idOlimpiada, idPais, idGenero
184     ):
185         try:
186             con = self.conectar()
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178         cursor = con.cursor()
179         cursor.execute(
180             "delete from resultados where
idOlimpiada = %s and idPais = %s and idGenero = %s",
181             (idOlimpiada, idPais, idGenero)
182         )
183         con.commit()
184         return cursor.rowcount > 0
185     except:
186         return False
187     finally:
188         self.desconectar()
189
190     def consultar(self, filter):
191         try:
192             con = self.conectar()
193             cursor = con.cursor()
194             cursor.execute(f"select * from
resultados where {filter}")
195             return cursor.fetchall()
196         except:
197             return []
198         finally:
199             self.desconectar()
200
201
202 # prueba
203 if __name__ == "__main__":
204     pais = sqlPais("localhost", "root", "password",
"olimpiadas")
205     print(pais.insertar(1, "mexico"))
206     print(pais.consultar("id = 1"))
207     print(pais.editar(1, "mexico editado"))
208     print(pais.eliminar(1))
209
210     olimpiada = sqlOlimpiada("localhost", "root", "
password", "olimpiadas")
211     print(olimpiada.insertar(1, 2024))
212     print(olimpiada.consultar("year_olimpiada > 2000
"))
213     print(olimpiada.editar(1, 2028))

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214     print(olimpiada.eliminar(1))
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